

The $\mathfrak{A}^{\dagger(1)} \times \mathfrak{A}^{(3)}$ row

Checking the split into N_1 (one ρ each) and N_2 (the two- ρ piece)

$$\ell_k \equiv \log \frac{k^+}{\Lambda}, \quad r \equiv x - w, \quad r_0 = \frac{2e^{-\gamma_E}}{\mu}, \quad b_0 = \frac{11N_c}{3}$$

Verdict. N_1 is **correct** — prefactor, transverse structure and both colour brackets all reproduce exactly (§1–2), and its logarithm bracket collapses to something much simpler than written (§3). N_2 's **prefactor is also correct**, but its ket colour bracket is **index-inconsistent**: f^{abc} cannot be pulled out as an overall factor, because the two terms carry genuinely different colour tensors (§4). Four smaller things need fixing (§5), one of which — the symbol $\mathbf{k}$ doing double duty as both the observed and the loop momentum — will bite. Two good signs along the way: $-\frac{67}{9} + \frac{\pi^2}{3}$ is the known NLO BFKL/BK constant, and the $\log \vee$ in $\mathfrak{A}^{(3)}$'s two- ρ bracket cancels **exactly**.

1. Prefactors

From $-\mathfrak{A}_i^a = -\frac{1}{\sqrt{2\pi}} \frac{ig}{\sqrt{k^+}} \int_x \frac{(x-z)^i}{(x-z)^2} \rho^a$, so $\mathfrak{A}^\dagger = -\frac{ig}{\sqrt{2\pi}\sqrt{k^+}}(\cdots)$, and the two pieces of $\mathfrak{A}^{(3)}$ carry $-\frac{ig^3 N_c}{16\sqrt{2\pi^3}\sqrt{k^+}}$ and $+\frac{ig^3}{32\sqrt{2\pi^5}\sqrt{k^+}}$:

piece	computed	quoted	ratio
N_1	$\frac{1}{(2\pi)^3} \frac{g^4 N_c}{32\pi^4} \frac{1}{k^+}$	same	1
N_2	$-\frac{1}{(2\pi)^3} \frac{g^4}{64\pi^6} \frac{1}{k^+}$	same (times f^{abc} , see §4)	1

Both exact. The i 's work out: $(-i) \times (-i) = -1$ for N_1 against the explicit minus in $\mathfrak{A}^{(3)}$'s first line, giving a real positive N_1 ; and $(-i) \times (+i) = +1$ against the explicit minus you carry in front of N_2 .

2. The colour brackets of N_1

$$\begin{aligned} \text{bra} &= -\bar{\mathfrak{A}}_i^{\dagger(1)a}(-w') + \mathfrak{A}_i^{\dagger(1)b'}(-w') U^{b'a}(w') = \frac{ig}{\sqrt{2\pi}\sqrt{k^+}} \int_{x'} \frac{(x' - w')^i}{(x' - w')^2} [U^{ab'}(x') - U^{b'a}(w')] \rho^{b'}(x'), \\ \text{ket} &= U^{ab}(w) \mathfrak{A}_i^{(3)b}(-w) - \bar{\mathfrak{A}}_i^{(3)a}(-w) \implies [U^{ab}(w) - U^{ab}(x)] \rho^b(x) \quad \checkmark \end{aligned}$$

The ket matches yours exactly. The bra matches *up to one transpose*: the row has $U^{b'a}(w')$, you wrote $U^{ab'}(w')$. These are different matrices ($\|U - U^T\| = 0.998$ on a random adjoint line), and only one ordering makes $[U(x') - U(w')]$ a sensible matrix difference. Pick one and use it in both places — the same issue appeared in the two- $\mathfrak{C}$ row. The transverse factor $\frac{(x'-w') \cdot (x-w)}{(x'-w')^2 (x-w)^2}$ and the measure $\int_{w, w', x, x'}$ are both right.

3. The logarithm bracket collapses

This is worth doing before anything else. The second factor in $\mathfrak{A}^{(3)}$ is

$$\frac{11}{3} + \log \frac{\Lambda}{k^+} + \log \left(\frac{\vee}{k^+} - 1 \right) - \log \frac{\vee}{\Lambda},$$

and the three logarithms combine (symbolic difference 0, and to machine precision numerically):

$$\log \frac{\Lambda}{k^+} + \log \left(\frac{\vee}{k^+} - 1 \right) - \log \frac{\vee}{\Lambda} = \log \frac{\Lambda^2(\vee - k^+)}{k^{+2}\vee} = \boxed{2 \log \frac{\Lambda}{k^+} + \log \left(1 - \frac{k^+}{\vee} \right)}$$

k^+	$\vee$	Λ	as written	collapsed
1.0	50	10^{-4}	-18.440883451	-18.440883451
2.3	17	10^{-6}	-29.442205212	-29.442205212
0.4	900	10^{-3}	-11.983373637	-11.983373637

So the second factor is simply $\frac{11}{3} - 2\ell_k + \log \left(1 - \frac{k^+}{\vee} \right)$.

3.1 And then all the $\vee$ dependence disappears

As $\vee \rightarrow \infty$: $\log(1 - \frac{k^+}{\vee}) \rightarrow 0$, and the tail $\log^2(1 - \frac{k^+}{\vee}) + \log(1 - \frac{k^+}{\vee}) \log \frac{\vee}{k^+} \rightarrow 0$ as well (the second because $\frac{k^+}{\vee} \log \vee \rightarrow 0$). N_1 therefore reduces to

$$\left[-\frac{2}{\epsilon} - 2\gamma_E - \log \frac{(x-w)^2 \mu^2}{4} \right] \left[\frac{11}{3} - 2 \log \frac{k^+}{\Lambda} \right] - \frac{67}{9} + \frac{\pi^2}{3}$$

Two checks fall out of this:

- $-\frac{67}{9} + \frac{\pi^2}{3} = -4.1545763$ is exactly the constant that appears in the NLO BFKL / BK kernel. Its presence here is a strong sign the $\mathfrak{A}^{(3)}$ expression is right.
- $\frac{11}{3} = b_0/N_c$ for pure glue, so the pole term is $N_c \cdot (-\frac{2}{\epsilon}) \cdot \frac{11}{3} = -\frac{2b_0}{\epsilon}$.

This $\frac{1}{\epsilon}$ is ultraviolet. Unlike the two- $\mathfrak{C}$ row and the $|\mathfrak{B}_2|^2$ row, where the pole came from $|z| \rightarrow \infty$ and was infrared, here it multiplies b_0 and **renormalises** g^2 . Concretely,

$$-\frac{2}{\epsilon} - 2\gamma_E - \log \frac{r^2 \mu^2}{4} = -2 \left[\frac{1}{\epsilon} + \log \frac{r}{r_0} \right], \quad r_0 = \frac{2e^{-\gamma_E}}{\mu},$$

which is exactly the position-space $\overline{\text{MS}}$ dictionary used earlier in this work. The residual $-2\ell_k$ multiplying the same bracket is a mixed $\frac{1}{\epsilon} \times$ rapidity-log term and belongs with the evolution counterterm, not with the coupling.

4. N_2 : f^{abc} cannot be pulled out front

$\mathfrak{A}^{(3)}$'s two- ρ piece carries $f^{abc}\{\rho^b(x), \rho^c(y)\}$, free index a . Building the ket bracket from that:

$$\text{term 1} = U^{ab}(w) \mathfrak{A}^{(3)b} = U^{ab}(w) f^{bcd}\{\rho^c(x), \rho^d(y)\}, \quad \text{term 2} = f^{abc} U^{bd}(x) U^{ce}(y) \{\rho^d(x), \rho^e(y)\}.$$

Term 2 is what you wrote and it is right. Term 1 is not: you wrote $f^{abc} U^{ab}(w) \{\rho^b(x), \rho^c(y)\}$, in which the index b appears **three times** — in f^{abc} , in U^{ab} and in ρ^b . In term 1 the f carries the index contracted with the Wilson line, not the free index, so it sits *inside*:

$$\left[U^{ab}(w) f^{bcd}\{\rho^c(x), \rho^d(y)\} - f^{abc} U^{bd}(x) U^{ce}(y) \{\rho^d(x), \rho^e(y)\} \right]$$

and the f^{abc} must come out of your prefactor $-\frac{1}{(2\pi)^3} \frac{g^4 f^{abc}}{64\pi^6 k^+}$, which becomes $-\frac{1}{(2\pi)^3} \frac{g^4}{64\pi^6 k^+}$.

The two tensors are genuinely different, not proportional — checked with explicit $SU(3)$ structure constants and random adjoint Wilson lines:

```
||U^{ab}(w) f^{bde}|| = 4.8990
||f^{abc} U^{bd}(x) U^{ce}(y)|| = 4.8990
|| difference || = 7.1113      -> not proportional
```

(They do coincide in structure when $x = y$, where adjoint invariance gives $f^{abc} U^{bd}(x) U^{ce}(x) = U^{aa'}(x) f^{a'de}$ to 2×10^{-16} — but $x \neq y$ here.)

4.1 One thing that works nicely in N_2

The $\log \vee$ in $\mathfrak{A}^{(3)}$'s momentum bracket **cancels exactly**. Collecting the coefficient of $\log \vee$ from all four $\vee$ -dependent terms:

$$\frac{(2p^2 - 2k \cdot p + k^2)k^i + k^2 p^i}{(k-p)^2} - \frac{p^2 k^i + k^2 p^i}{(k-p)^2} - k^i - 2k^i + 2k^i = \frac{(k-p)^2 k^i}{(k-p)^2} - k^i = 0.$$

The first two combine because $2p^2 - 2k \cdot p + k^2 - p^2 = (k-p)^2$, and the two $-2k^i$ logs cancel because $\log \frac{k^+ p^2}{\vee k^2 + k^+ p^2} \rightarrow -\log \frac{\vee k^2}{k^+ p^2}$. Symbolically exactly zero. So that piece is $\vee$ -stable, as it should be.

5. Four smaller fixes

1. **N_2 's measure omits x' .** It reads $\int_{w,w'} \cdots \int_{x,y}$, but x' appears both in $\frac{(x'-w')^i}{(x'-w')^2}$ and in $\rho^{b'}(x')$. It should be $\int_{x,y,x'}$.
2. **$\mathbf{k}$ means two different things in N_2 .** It is the observed transverse momentum in $e^{-i\mathbf{k}\cdot(w'-w)}$ and in $\frac{d^3N}{d^3k}$, and simultaneously the loop momentum in $\int d^2\mathbf{k}$ and $e^{i\mathbf{k}\cdot(w-y)}$. Rename the loop momentum (say $\mathbf{q}$) before you do anything with the w dependence, or the two will be combined by accident — they both multiply w in an exponent.
3. **The row is not real as it stands.** It is $[\mathfrak{A}^{\dagger(1)}] \times [\mathfrak{A}^{(3)}]$ in one ordering only, and the bracket $\mathcal{G}$ is attached to $(x-w)$ but not to $(x'-w')$, so the expression is complex. The physical result needs $+ \text{c.c.}$ (equivalently 2Re), or the companion row $\mathfrak{A}^{(3)\dagger} \times \mathfrak{A}^{(1)}$ written out separately.
4. **Subscripts.** The definitions write $\mathfrak{A}_{2i}^a$ and $\mathfrak{A}_{2i}^{(3)a}$ while the row uses $\mathfrak{A}_i^{(1)}$ and $\mathfrak{A}_i^{(3)}$. Same object — just make it uniform.

6. N_1 , corrected and simplified

$$\begin{aligned} \frac{d^3 N_1}{d^3 k} &= \frac{1}{(2\pi)^3} \frac{g^4 N_c}{32\pi^4} \frac{1}{k^+} \int_{w,w',x,x'} e^{-ik\cdot(w'-w)} \frac{(x'-w') \cdot (x-w)}{(x'-w')^2 (x-w)^2} \\ &\times \left\{ -2 \left[\frac{1}{\epsilon} + \log \frac{|x-w|}{r_0} \right] \left[\frac{11}{3} - 2 \log \frac{k^+}{\Lambda} + \log \left(1 - \frac{k^+}{\sqrt{v}} \right) \right] - \frac{67}{9} + \frac{\pi^2}{3} + \log^2 \left(1 - \frac{k^+}{\sqrt{v}} \right) + \log \left(1 - \frac{k^+}{\sqrt{v}} \right) \log \frac{\sqrt{v}}{k^+} \right\} \\ &\times \left[U^{ab'}(x') - U^{ab'}(w') \right] \rho^{b'}(x') \left[U^{ab}(w) - U^{ab}(x) \right] \rho^b(x) + \text{c.c.} \end{aligned}$$

$r_0 = 2e^{-\gamma_E}/\mu$. Everything in the second $\sqrt{}$ -bracket vanishes as $\sqrt{} \rightarrow \infty$, leaving $-2 \left[\frac{1}{\epsilon} + \log \frac{|x-w|}{r_0} \right] \left[\frac{11}{3} - 2\ell_k \right] - \frac{67}{9} + \frac{\pi^2}{3}$.

zint/src/row10_check.py (§1,3,4,1), row10_colour.py (§2,4,5) — every number above.